Primate Diet Transcriptomics

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Bioinformatics

Agenda

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02 Dataset Description

03 Data Behaviour

04 Getting the DEGs

05 PCA

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Research Objectives

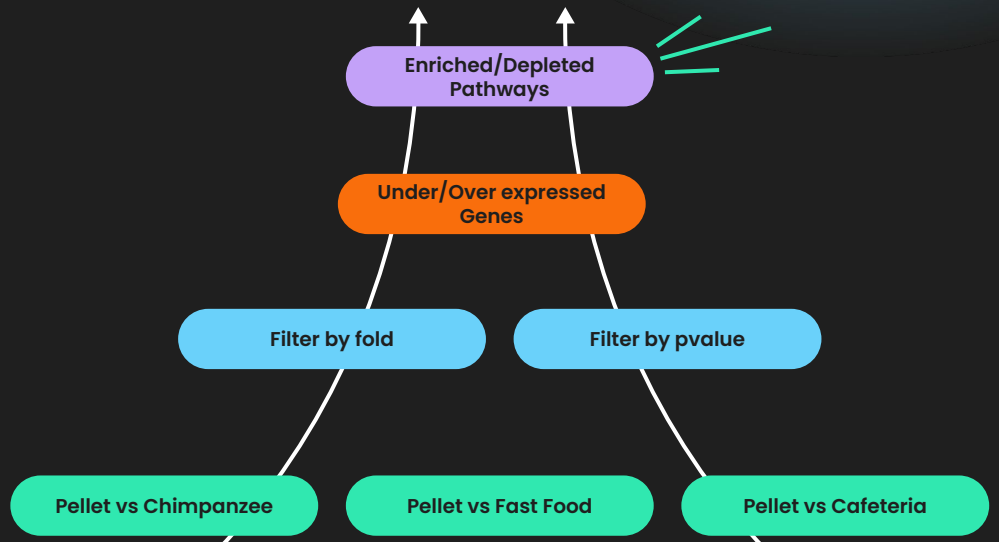
Research objectives

Compare gene expression in mice fed:

- Chimpanzee diet
- Human Cafeteria diet
- Fast Food diet

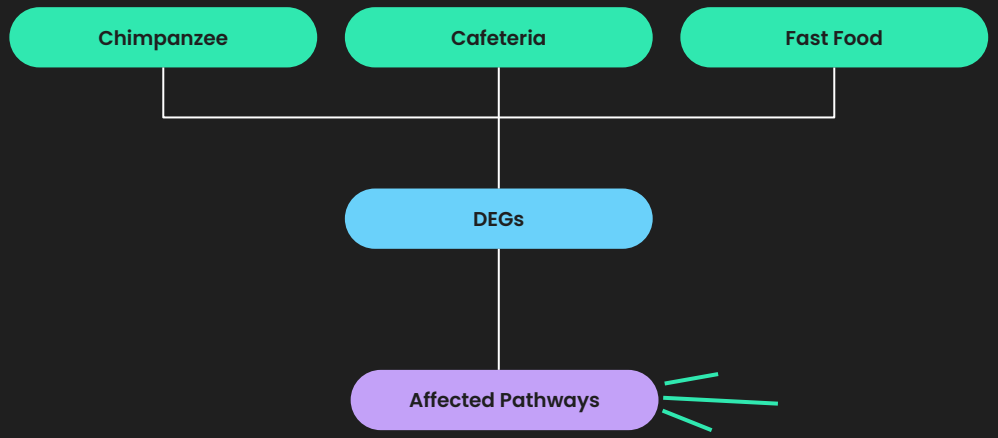
With mice fed a pellet diet (control)

For each condition, identify the differentially expressed genes and the biological pathways they contribute to



Research objectives

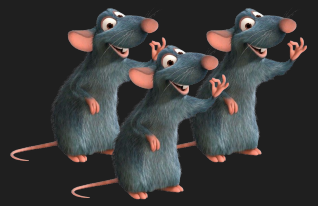
Identify the genes that contribute to the variation between the experimental groups only through **PCA** and the affected pathways



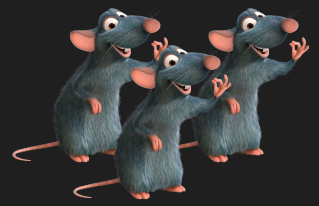
Dataset Description

Experimental Setup

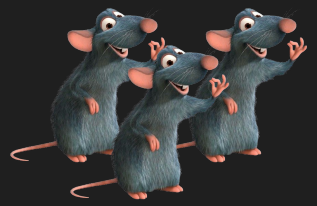
Batch 1



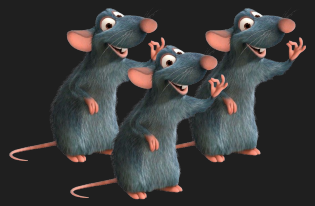
Pellet Diet



Chimpanzee Diet

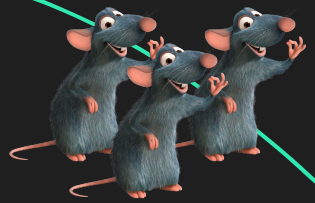


Cafeteria Diet



Fast Food Diet

Batch 2



Experimental Setup

Batch 1

Pellet Diet

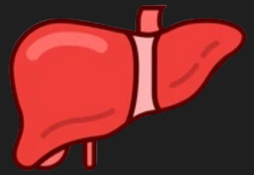
Chimpanzee Diet

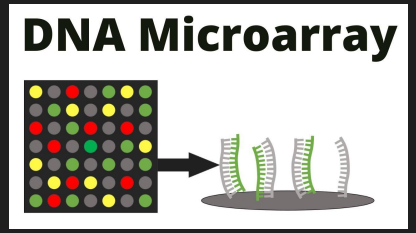
Cafeteria Diet

Fast Food Diet

Batch 2

Experimental Setup





Dataset

3 mice * 2 batches * 4 diets = 24 samples
43014 genes

```
[ ]  1  dim(data)
```

43014 · 24

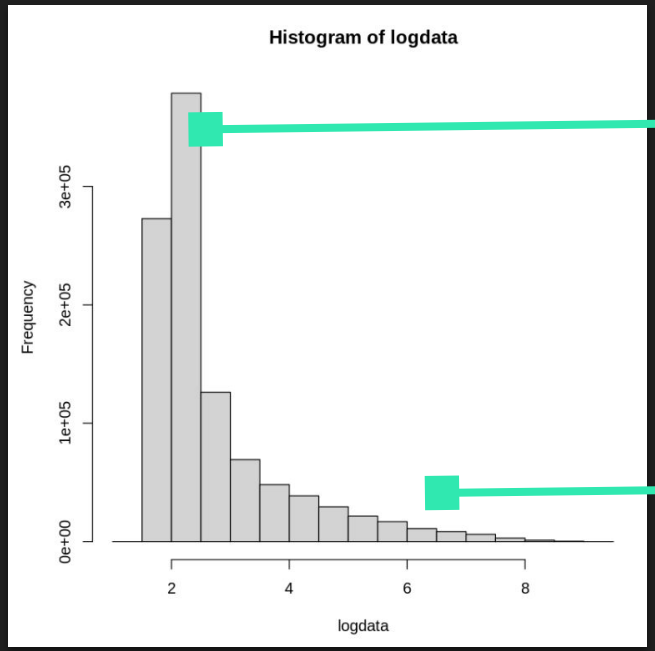
```
1  head (data)
```

A matrix: 6 × 24 of type int

	ChimpanzeeDiet1.1	ChimpanzeeDiet1.2	ChimpanzeeDiet1.3	ChimpanzeeDiet2.1	ChimpanzeeDiet2.2	ChimpanzeeDiet2.3	FastFoodDiet1.1	FastFoodDiet1.2	FastFoodDiet1.3	FastFoodDiet2.1	···	HumanCafeDiet1.3	HumanCafeDiet2.1	HumanCafeDiet2.2	HumanCafeDiet2.3	PelletDiet1.1	PelletDiet1.2	PelletDiet1.3	PelletDiet2.1	PelletDiet2.2	PelletDiet2.3
1415670_at	57	73	83	33	24	37	60	65	59	40	···	73	39	38	23	97	72	88	58	62	71
1415671_at	283	313	359	200	187	237	322	278	332	296	···	318	276	240	172	342	332	348	269	267	294
1415672_at	233	213	205	142	162	216	278	274	372	274	···	323	216	284	153	349	271	297	226	233	298
1415673_at	13	13	13	15	12	15	16	13	13	14	···	18	12	13	12	19	14	17	14	14	16
1415674_a_at	56	58	67	35	37	50	78	74	71	51	···	58	60	52	37	97	80	77	55	68	78
1415675_at	43	45	46	26	36	28	45	45	40	31	···	37	47	34	30	40	42	47	41	40	35

Data Behaviour

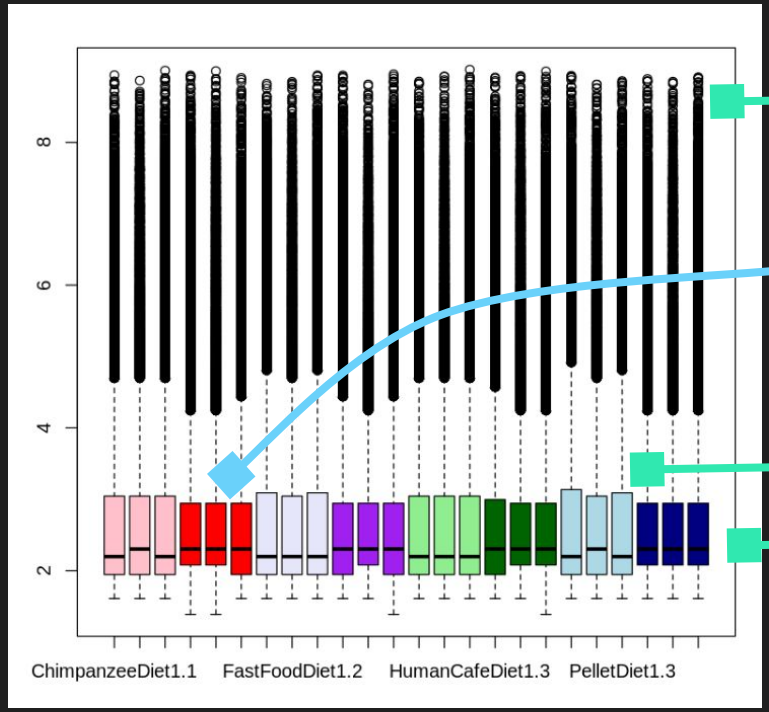
Data Behaviour



Single Peak

Positively Skewed

Data Behaviour



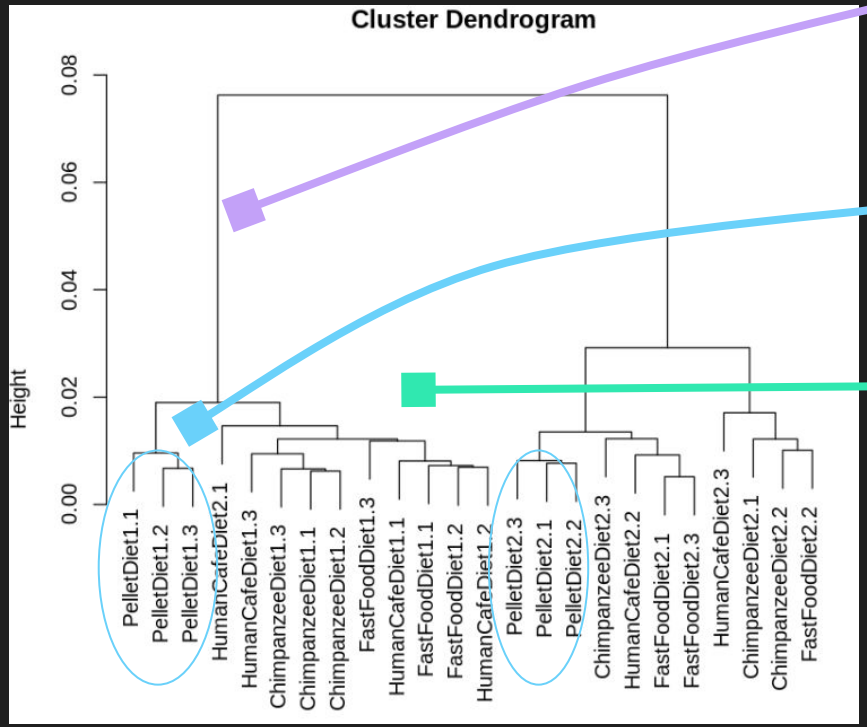
Similar Whiskers

Smaller IQRs in Batch 2

Similar IQRs

Consistent Medians

Data Behaviour



Every batch clusters together → strong batch effect

Pellet Diet forms a tight cluster → high similarity → good control

Other diets form less distinct clusters, especially fast food and cafeteria

Getting the DEGs

Step 1: Filter by Fold

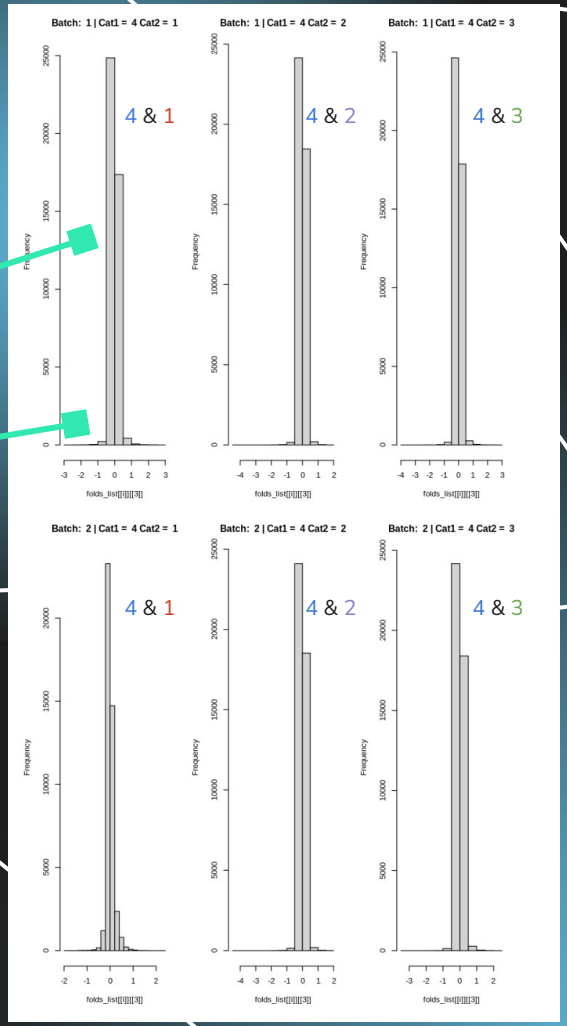
Biological Significance

Mostly around 0

Very few abs > 1

Key:

1 = Chimpanzee diet
2 = Fast Food diet
3 = Human Cafeteria diet
4 = Pellet diet



Cutoff = 1.0

Step 2: Filter by p-value

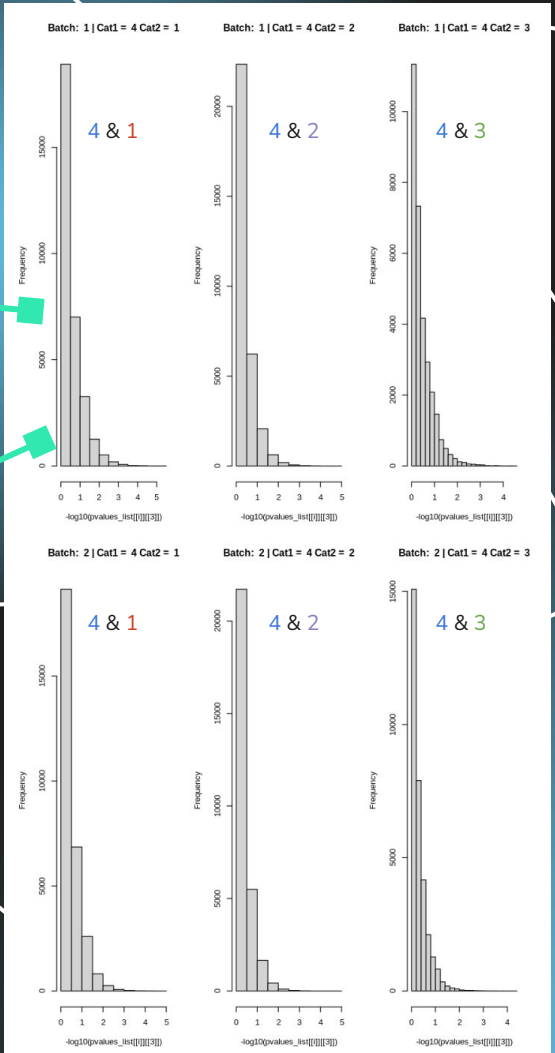
Statistical Significance

Most at 0

Positively skewed

Key:

1 = Chimpanzee diet
2 = Fast Food diet
3 = Human Cafeteria diet
4 = Pellet diet



Cutoff = 0.05

Volcano Plots

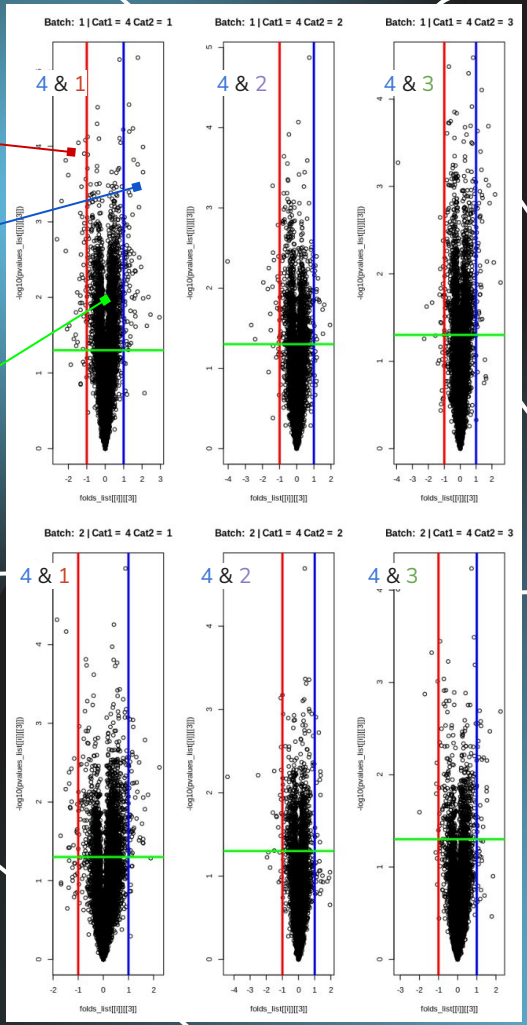
Both filters

Key:
1 = Chimpanzee diet
2 = Fast Food diet
3 = Human Cafeteria diet
4 = Pellet diet

Underexpressed

Overexpressed

Statistically significant



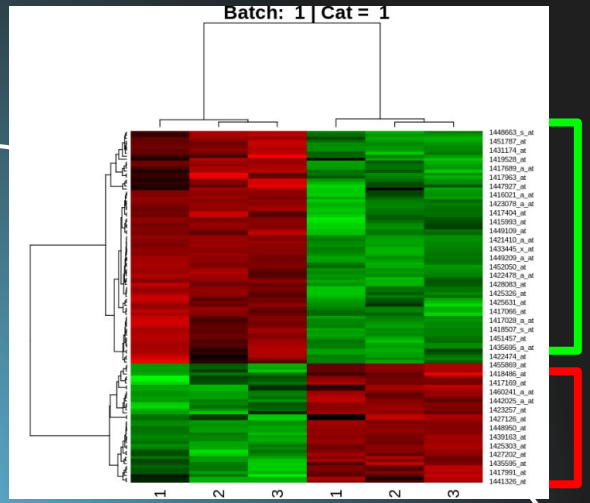
Fold cutoff = 1
p-value Cutoff = 0.05

Differentially Expressed Genes

Chimpanzee Diet

Batch 1

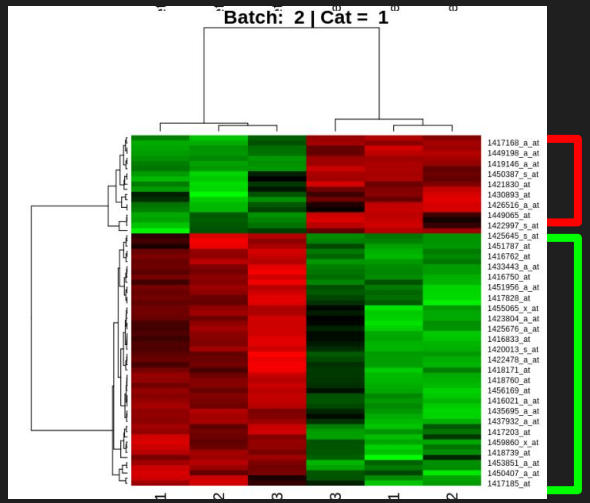
118 DEGs



pellet Chimpanzee

Batch 2

68 DEGs



pellet Chimpanzee

Union

139 DEGs

94 overexpressed

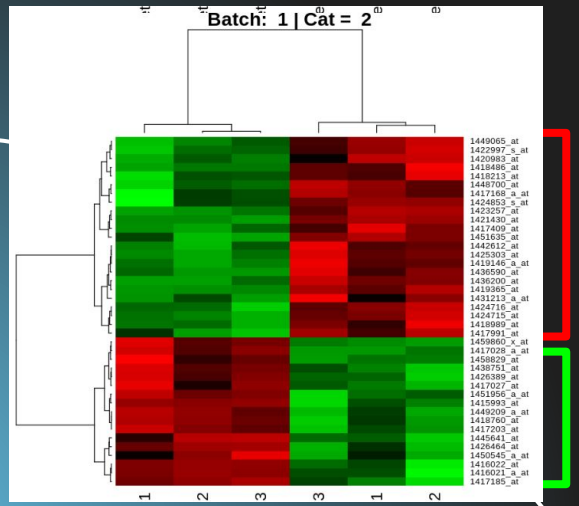
45 underexpressed

Differentially Expressed Genes

Fast Food Diet

Batch 1

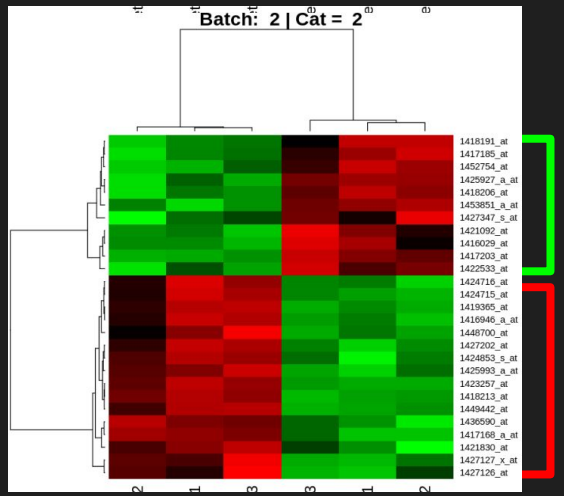
40 DEGs



pellet Fast Food

Batch 2

27 DEGs



Fast Food pellet

Union

56 DEGs

26 overexpressed

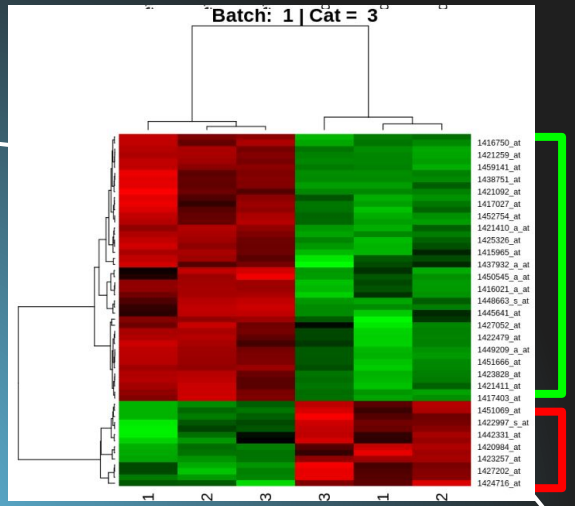
30 underexpressed

Differentially Expressed Genes

Human Cafeteria Diet

Batch 1

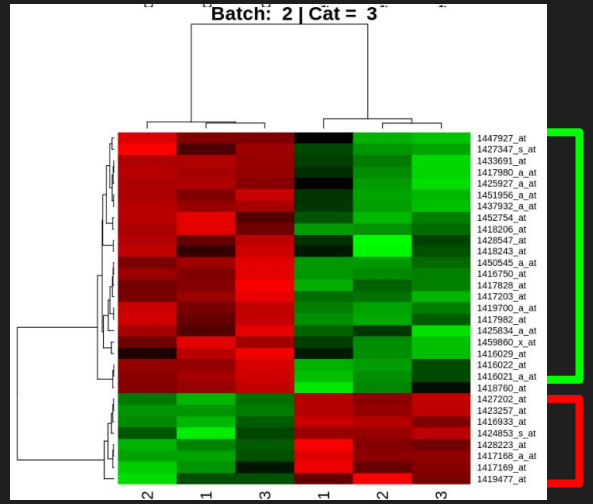
58 DEGs



pellet Cafeteria

Batch 2

31 DEGs



pellet Cafeteria

Union

74 DEGs

55 overexpressed

19 underexpressed

PCA

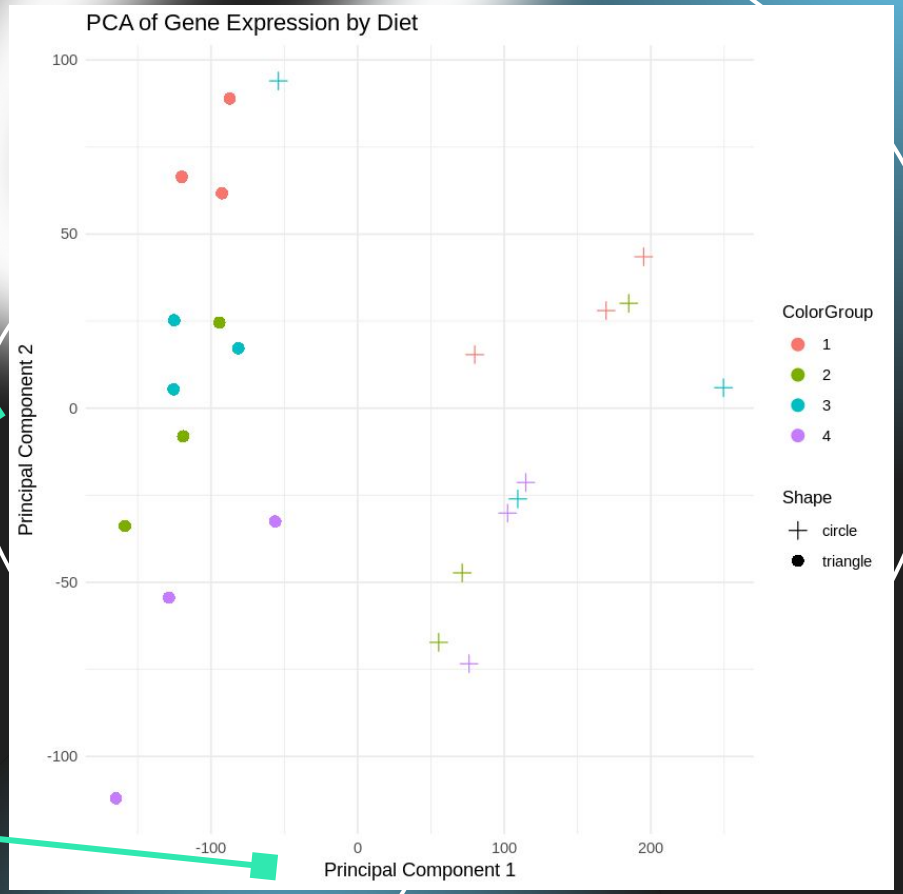
PCA of Both Batches

Key:
1 = Chimpanzee diet
2 = Fast Food diet
3 = Human Cafeteria diet
4 = Pellet diet

PC2 separates by diet

83 genes in PC2 that are 3 standard deviations above the mean

PC1 separates by batch

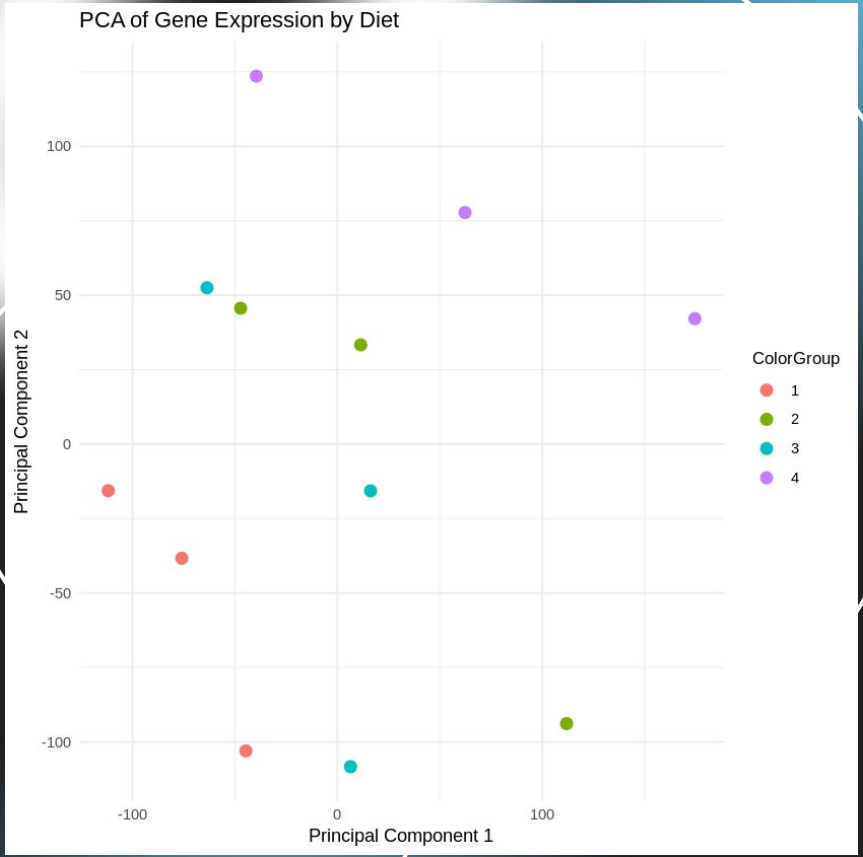


PCA of Batch 1

Key:
1 = Chimpanzee diet
2 = Fast Food diet
3 = Human Cafeteria diet
4 = Pellet diet

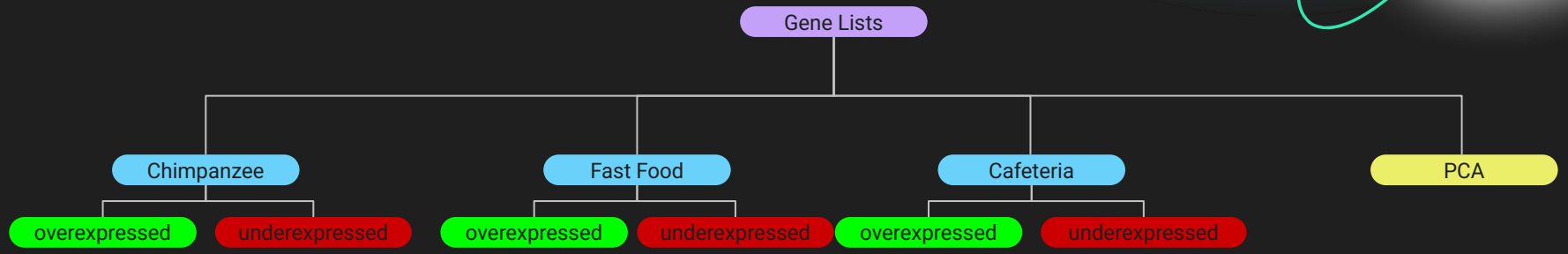
37 genes in PC1 and PC2 that are 1 standard deviation above the mean

All of them were also obtained through the first PCA (no new genes)



66 PCA genes not in DEGs → contribute the most to variation between experimental groups

Enrichment Analysis



PANTHER
(GO database)

Overrepresented Pathways

Chimpanzee Diet 🥬🍌 – Upregulated

Enriched Pathways:

1. Cholesterol Biosynthesis
2. Sterol/Cholesterol/Steroid Metabolism
3. Fatty Acid Elongation & Lipid Assembly
4. Alcohol Metabolism
5. General Biosynthetic & Metabolic Pathways

Biological Interpretation:

1. Mice fed a chimpanzee-style diet (low cholesterol, high fiber, plant-based) showed upregulation of lipid/sterol synthesis genes.
2. This indicates a compensatory response to a diet deficient in cholesterol and long-chain fatty acids.
3. Regulatory networks drive this response.

Chimpanzee Diet 🥬🍌 - Downregulated

Enriched Pathways:

1. Vitamin B3 (Nicotinamide) and NAD(P) Metabolism
2. Purine/Nucleotide Catabolism
3. Long-Chain Fatty Acid Oxidation
4. General Small Molecule & Organic Acid Metabolism

Biological Interpretation:

1. The liver reduces energy-demanding catabolic pathways due to a less lipid-rich, more fiber-heavy diet.
2. Less need for internal vitamin and nucleotide generation because of increased external supply from the plant-rich diet.

Fast Food Diet 🍔 – Upregulated

Enriched Pathways & biological impacts

1. Lipid & Sterol Biosynthesis / Transport
   - Increased cholesterol and fatty acid synthesis in response to high dietary fat
   - Enhanced bile acid production and lipid droplet formation indicating fat accumulation in the liver
2. Neuroimmune & Brain Barrier Processes
   - Activation of microglia and astrocytes leading to low-grade neuroinflammation
   - Altered endocannabinoid and synaptic signaling affecting brain function and reward pathways
   - Changes in blood–brain barrier lipid transport mechanisms

Fast Food Diet 🍔 - Upregulated

Enriched Pathways & biological impacts

3. Systemic Immune and Inflammatory Responses
    - Increased immune system activation
    - Elevated oxidative stress
4. Circadian Rhythm and Metabolic Hormone Regulation
    - Disruption of metabolism and energy balance
    - Altered leptin response indicating possible leptin resistance and obesity risk

Fast Food Diet 🍔 - Downregulated

Enriched Pathways:

1. Fatty Acid/Lipid Catabolism
2. Purine & Nucleotide Catabolism
3. Monocarboxylic, Carboxylic, Oxoacid & Organic Acid Metabolism

Biological Interpretation:

1. Fat builds up in the liver because the liver slows down breaking down fats.
2. The liver's energy use changes — it relies less on burning fat and more on other processes.
3. Balance of important molecules changes, which can affect how liver cells handle stress.
4. Waste products and acids may build up, causing extra stress in the liver.
5. This can lead to liver inflammation and damage over time, making the liver unhealthy.

Human Cafeteria Diet 🥪 – Upregulated

Enriched Pathways & biological impacts:

1. Acetyl-CoA and Acyl-CoA Metabolism
   - Liver cells strongly increase production of activated fatty acid building blocks.
2. Sterol and Cholesterol Biosynthesis
   - Despite dietary cholesterol, the liver boosts its own cholesterol production, likely due to high sugar and fat intake.
3. Fatty Acid and Lipid Biosynthesis
   - The liver is actively making fatty acids and complex lipids. This supports fat storage and lipid transport.
4. Organic Acid and Small Molecule Biosynthesis

Human Cafeteria Diet 🥪 – Downregulated

Enriched Pathway & biological impacts:

1. Long-Chain Fatty Acid Metabolism
    - Genes responsible for burning (oxidizing) long-chain fatty acids (C16 and longer) in the mitochondria are turned down. This reduces the liver's ability to break down dietary fats.

Only 19 genes were significantly underexpressed → not many affected pathways.

PCA Genes

Enriched Pathways:

1. Lipid & Sterol Biosynthesis / Regulation
2. Fatty-Acid Activation & Thioester Metabolism
3. Small Molecule & Organic Acid Metabolism
4. Regulation of Anabolic vs. Catabolic Programs

Some key Enriched GO Terms:

1. Small molecule metabolic process (GO:0044281)
2. Small molecule biosynthetic process (GO:0044283)
3. Carboxylic acid metabolic process (GO:0019752)
4. Oxoacid metabolic process (GO:0043436)
5. Organic acid metabolic process (GO:0006082)
6. Purine-containing compound metabolic process (GO:0072521)
7. Organophosphate metabolic process (GO:0019637)
8. Acyl-CoA metabolic process (GO:0006637)
9. Fatty acid metabolic process (GO:0006631)
10. Sulfur compound metabolic process (GO:0006790)

In Conclusion

Different diets affect gene expression in the liver, which activates/inhibits different metabolic pathways to adapt with the difference in nutrient intake

Thank you!

Questions?